GIS for Non-Majors Final Project Report: An Analysis of Public Shelter Availability in Tornado Prone Regions

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Topic, Issue, and Problem

My project, "An Analysis of Public Shelter Availability in Tornado Prone Regions," highlights a major gap in public safety within the most dangerous tornado corridors of the US.

The core issue I wanted to visualize is the disconnect between tornado warning lead times and the actual ability of residents to reach safety.

NWS data confirms that severe (EF3+) tornadoes typically come with a 16-minute warning, with most of these responsible for the vast majority of the 384 deaths and over $8 billion in property damage in my dataset.

My analysis shows that a staggering 67.2% of the population in the highest-risk zone lives more than a 16-minute drive from a public shelter.

This map serves as a guide for emergency managers, identifying exactly where new shelters are needed most to protect the 29.5 million people currently living in these un-served "tornado shelter deserts."

GIS Skills Used

1. **Using Raster Data (Spatial Analysis)**

   To define my study area objectively, I didn't just pick states; I let the data decide. I used the Kernel Density tool on historical NOAA tornado reports to create a raster heat map of tornado intensity. By weighting the density with the EF-Scale, I ensured that violent EF4 and EF5 tornadoes influenced the map more than weaker EF3 ones. I then reclassified this raster to isolate the top tier of risk values (density > 0.00009) and converted that high-risk raster area into a polygon. This gave me a scientifically defined "Hotspot Zone" to focus my study on.

2. **Geoprocessing Tools**

   To determine who could actually reach a shelter, I used a chain of geoprocessing tools. Instead of a simple circle buffer, I used the Service Area tool (from the Network Analyst extension) to calculate a realistic 16-minute drive-time polygon around every shelter. Because I had over 2,000 shelter points, I hit a processing limit, so I had to use the Select Layer by Attribute tool to split the data into three batches. After running the service area analysis on each batch, I used the Merge tool to combine them and the Dissolve tool to melt them into one clean "Served_Area" layer.

3. **Tables and Joins**

   To get accurate population numbers, I used Census Block Groups, which are the smallest units that have the detailed data I needed. I downloaded the spatial shapefiles and the population tables of the latest 2022 US Census. To avoid the data errors that come from clipping census polygons, I used Select By Location with the "Have their center in" relationship. This allowed me to select only the block groups inside my hotspot zone and then identify which of those were outside the served area to get my final population count.

Discussion and Limitations

The final output provides a clear, but complex, picture of public safety. My analysis has two main caveats that need to be understood when applying the results:

- Data Reliability: The list of public shelters came from findyourtornadoshelter.com, a robust but community-driven source. While it's the best available inventory, any positional inaccuracies or missing shelters could slightly shift the boundaries of my 16-minute service area.

- The Urban-Rural Paradox: The calculated 67.2% Un-Served rate is not uniform. Cities like Dallas and Atlanta show poor formal public shelter coverage in their centers. However, these major cities are likely safer than the data suggests because urban buildings and homes often provide shelter (basements, reinforced cores) that isn't included in my public inventory. The true public safety crisis is concentrated in vulnerable rural and exurban areas—communities that lack basements, are home to mobile parks, and are far from any reinforced structure. My map is therefore best used to identify these underserved rural populations where new, dedicated public shelters are most desperately needed.